MORPHISMS OF AUGMENTED SIMPLICIAL SITES GIVE COMMUTATIVE DIAGRAMS OF TOTAL AND BASE TOPOI

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: this is the closing lemma applied by the comparison diagrams of cards L46173 (fppf-etale) and L46174 (ph-etale); the componentwise reduction uses lemma-describe-sheaves-simplicial-site-site; no verification is omitted.

Editorial note. The label lemma-morphism-augmentation-simplicial-sites names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The morphism of augmented simplicial sites setup. Same chapter, Remark \label{remark-morphism-augmentation-simplicial-sites}. Statement lines 1403-1410: the data C_n, f_phi, u_phi, D, a_0, C'_n, f'_phi, u'_phi, D', a'_0 and morphisms h_n, n >= -1, as in the remark.
- The total topos morphism. Same chapter, \label{lemma-morphism-simplicial-sites} (card L46154). Proof lines 1419-1420: the morphism h (with component h_n) is defined by this lemma.
- The augmentation topos morphisms. Same chapter, \label{lemma-augmentation-site}. Proof lines 1420-1423: the morphisms a and a' are defined by this lemma; the only remaining content is the commutativity of the diagram.
- The componentwise description of the total topoi. Same chapter, \label{lemma-describe-sheaves-simplicial-site-site}. Proof lines 1425-1430: by the component diagrams of the morphism lemma and the componentwise description of the total topoi, it suffices that the component squares commute for all n.

Source reference. Original reference 0D70: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1141–1141. Prerequisite pointer invoked at proof line(s) 1420; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 953–953. Prerequisite pointer invoked at proof

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line(s) 1428; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D96: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1273–1273. Prerequisite pointer invoked at proof line(s) 1419; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D98: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1372–1372. Prerequisite pointer invoked at proof line(s) 1406; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}_n, f_\varphi, u_\varphi, \mathcal{D}, a_0$, $\mathcal{C}'_n, f'_\varphi, u'_\varphi, \mathcal{D}', a'_0$, and h_n, $n \geq -1$ be as in Remark 0D98. Then we obtain a commutative diagram*

$$\begin{array}{ccc} Sh(\mathcal{C}_{total}) & \xrightarrow{h_{total}} & Sh(\mathcal{C}'_{total}) \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle a'} \\ Sh(\mathcal{D}) & \xrightarrow{h_{-1}} & Sh(\mathcal{D}') \end{array}$$

Proof. The morphism h is defined in Lemma 0D96. The morphisms a and a' are defined in Lemma 0D70. Thus the only thing is to prove the commutativity of the diagram. To do this, we prove that $a^{-1} \circ h_{-1}^{-1} = h_{total}^{-1} \circ (a')^{-1}$. By the commutative diagrams of Lemma 0D96 and the description of $Sh(\mathcal{C}_{total})$ and $Sh(\mathcal{C}'_{total})$ in terms of components in Lemma 09WF, it suffices to show that

$$\begin{array}{ccc} Sh(\mathcal{C}_n) & \xrightarrow{h_n} & Sh(\mathcal{C}'_n) \\ {\scriptstyle a_n}\downarrow & & \downarrow{\scriptstyle a'_n} \\ Sh(\mathcal{D}) & \xrightarrow{h_{-1}} & Sh(\mathcal{D}') \end{array}$$

commutes for all n. This follows from the case for $n = 0$ (which is an assumption in Remark 0D98) and for $n > 0$ we pick $\varphi : [0] \to [n]$ and then the required commutativity follows from the case $n = 0$ and the relations $a_n = a_0 \circ f_\varphi$ and $a'_n = a'_0 \circ f'_\varphi$ as well as the commutation relations $f'_\varphi \circ h_n = h_0 \circ f_\varphi$. $\square$